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Enantioselective synthesis of mixed 3,3′-bisindoles via a phosphine-catalyzed umpolung γ-addition of 3′-indolyl-3-oxindoles to allenoates

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ABSTRACT

An enantioselective umpolung γ-addition reaction of 3′-indolyl-3-oxindoles to allenoates catalyzed by amino acid-derived bifunctional phosphine catalysts has been developed. A wide range of chiral mixed 3,3′-bisindole scaffolds containing an all-carbon quaternary stereogenic center were obtained in high yields and with excellent enantioselectivities. 3,3′-Bisindoles are valuable synthetic intermediates, the employment of which led to formal total syntheses of (+)-Chimonanthine, (+)-Folicanthine and (−)-Calycanthine, as well as facile creation of useful pyrrolidinoindoline core structure.

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1. Introduction

The 3,3′-bisindole motif is an important structural unit that is widely present in hexahydropyrroloindole (HPI) alkaloids (Fig. 1), a family of natural products possessing fascinating molecular architectures and promising biological activity profiles [1–7]. Efficient total syntheses of HPIs often rely on tactical synthetic manipulations of key precursors/intermediates consisting of mixed 3,3′-bisindole cores [8–14]. From a synthetic viewpoint, preparation of optically enriched mixed 3,3′-bisindole structures is a long-standing and challenging task, especially in view of the presence of an all-carbon quaternary stereogenic center at the indole C_3-position [15–20]. In recent years, various synthetic strategies have been developed for their asymmetric synthesis, including: 1,4-conjugate additions or epoxide-opening via Friedel-Crafts reactions of indoles [21–24], Rh- or Ru-mediated multi-component tandem reactions [25–27], as well as nucleophilic substitutions of various 3-indolylmethanols and their structural analogues [28–36]. Undoubtedly, given the extreme synthetic value of the mixed 3,3′-bisindole motifs, there still exists an urgent need to devise efficient approaches for their enantioselective synthesis, especially those methods amenable for the preparation of a broad spectrum of natural products.

To construct mixed 3,3′-bisindoles containing a quaternary stereogenic center, we envisioned that 3′-indolyl-3-oxindoles may be utilized as a starting material. In contrast to most aforementioned literature approaches employing an advanced bisindole intermediate containing a leaving group at the quaternary center, there were only two reports to date using simple 3′-indolyl-3-oxindole substrates via transition metal catalysis; Trost et al. [37] disclosed a palladium-catalyzed asymmetric allylic alkylation with allenes, and Arai et al. [38] documented a nickel-mediated 1,4-conjugate addition to nitroethylene. We envisaged that the employment of prochiral pronucleophilic 3′-indolyl-3-oxindoles in a phosphine-mediated catalytic umpolung γ-addition reaction may offer brand new synthetic perspectives – inversion of reactivities of substrates is especially noteworthy. The past decade has seen marvelous advancement of asymmetric phosphine catalysis [39–45], and in these endeavors, our group introduced a new family of amino acid-based bifunctional phosphines and applied them to a wide range of asymmetric transformations [46–53]. Considered to be one of the landmark discoveries in phosphine catalysis, Trost and Lu' disclosure [54,55] of phosphine-catalyzed umpolung γ-addition offers a new dimension in organic synthesis, and not surprisingly, many excellent enantioselective umpolung γ-addition reactions have been uncovered over the time [56–68]. Among them, our group and the Zhao group [61,64,69] had demonstrated that 3,3′-disubstituted oxindoles could be easily accessed by employing phosphine-catalyzed umpolung γ-addition reaction.

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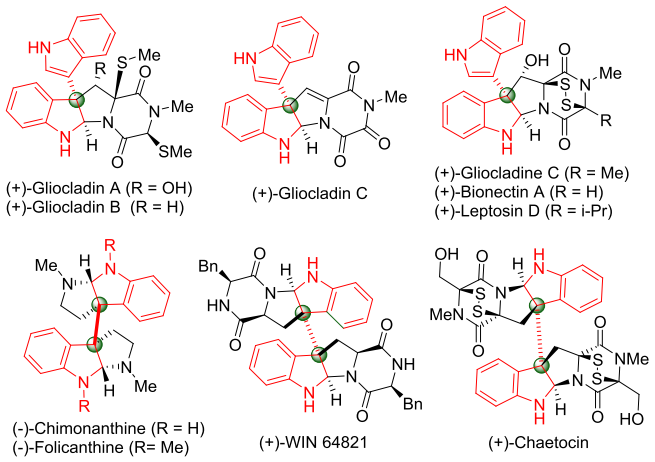


Fig. 1. (Color online) Representative HPI alkaloids containing 3,3′-bisindole core structures and all-carbon quaternary stereogenic center(s).

In recent years, we have developed keen interests in devising new catalytic methods for asymmetric construction of quaternary or tetra-substituted stereogenic carbon centers [70–74]. Our proposed strategy for the creation of chiral mixed 3,3′-bisindole scaffolds bearing a quaternary center via a phosphine-catalyzed γ-addition is illustrated in Scheme 1. The 3′-indolyl-3-oxindoles appear to be ideal prochiral pronucleophiles, the hydrogen atom at the oxindole 3-position is sufficiently acidic for basic abstraction, thus allowing for subsequent reactions to take place. The reactivity inversion of substrates is appealing: 3′-indolyl-3-oxindoles serve as a nucleophilic partner, as opposed to many existing methods [28–36] employing leaving group-bearing 3,3′-bisindoles as an electrophile. Moreover, in addition to allenoates, other electrophilic partners could also be utilized, therefore greatly broadening the reaction scope, making a diverse range of target molecules synthetically accessible. The employment of 3′-indolyl-3-oxindoles with no leaving group also makes this approach more attractive in a "green" perspective – the envisaged γ-addition is an atom-economical process [75,76]. Herein, we document a highly enantioselective umpolung γ-addition reaction of 3′-indolyl-3-oxindoles to allenoates, for the effective creation of chiral mixed 3,3′-bisindole scaffolds, which are versatile synthetic intermediates for the total syntheses of HPI alkaloids.

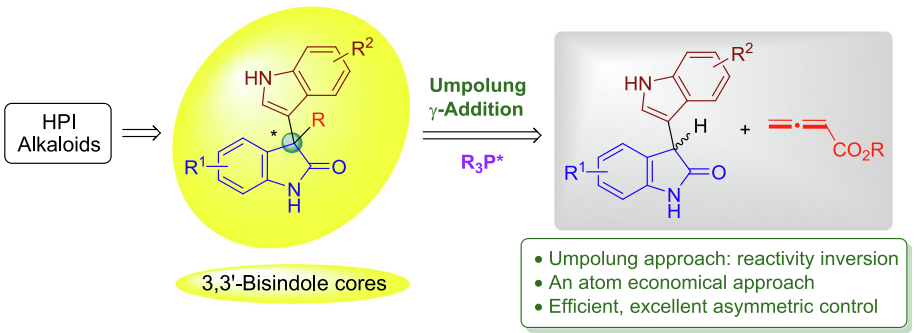
2. Experimental

All chemicals which are commercially available were used without further purification. Thin-layer chromatography (TLC) was performed on silica gel plates (60F-254) using UV-light (254 and 365 nm). Flash chromatography was conducted on silica gel (200–300 mesh). ^{1}H and ^{13}C NMR spectra were recorded at ambient temperature in either $CDCl_3$ or CD_3CN or CD_2Cl_2 on a Bruker AMX500 (500 MHz) or AMX400 (400 MHz) spectrometer. Chemical shifts were reported in parts per million (ppm). The data are reported as follows: for ^{1}H NMR, chemical shift in ppm from tetramethylsilane with the solvent as internal standard ($CDCl_3$ δ 7.26 ppm, CD_3CN δ 1.94 ppm, CD_2Cl_2 δ 5.32 ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet or overlap of non-equivalent resonances), integration; for ^{13}C NMR, chemical shift in ppm from tetramethylsilane with the solvent as internal indicator ($CDCl_3$ δ 77.16 ppm, CD_3CN δ 1.32–118.26 ppm, CD_2Cl_2 δ 53.84 ppm), multiplicity with respect to protons . All high-resolution mass spectra were performed on a Finnigan/MAT 95XL-T spectrometer to be given in *m/z*. Optical rotations were measured using an Anton Paar MCP-100 digital polarimeter. Enantiomeric excesses were determined by HPLC analysis on a chiral stationary phase using CHIRALPAK® IE column eluting with hexane/isopropanol mixtures as indicated.

Representative procedure for **P1**-catalyzed umpolung γ-addition of 3′-indolyl-3-oxindoles to allenoates: to a stirring anhydrous toluene (1 mL) solution of 3,3′-indolyloxindoles **1** (1 equiv., 0.1 mmol) and catalyst **P1** (0.2 equiv. 0.02 mmol, 9.5 mg) in an oven-dried reaction tube with a magnetic stir bar at the indicated temperature was added allenoates **2** (1.2 equiv., 0.12 mmol). The reaction mixture was stirred until the completion of the reaction (2–48 h, monitored by thin-layer chromatography TLC). The solvent was then removed under reduced pressure and the residue was purified by flash column chromatography on silica gel, after the pretreatment of the silica with Et_3N (hexane:dichloro methane = 5:1 to 3:1 with 2% Et_3N) to afford the product **3a**–**v**.

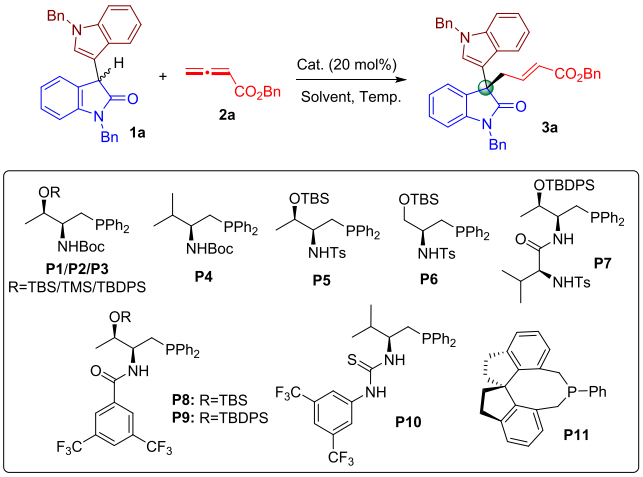
3. Results and discussion

To initiate our investigation, the γ-addition reaction of 3′-indolyl-3-oxindole **1a** to allenoate **2a** was chosen as a model reaction, and catalytic effects of a range of phosphines were examined (Table 1). L-Thr or L-Val-derived phosphine-carbamate catalysts (**P1** to **P4**) displayed very good catalytic effects, furnishing the desired γ-addition product **3a** in excellent yields and high enantiomeric excesses (Table 1, entries 1–4). L-Thr-derived **P5** bearing a sulfonamide moiety was also found to be a good catalyst (Table 1, entry 5), whereas the analogous L-Ser-derived **P6** was a poor catalyst (Table 1, entry 6). Subsequent examinations showed that dipeptidic phosphine-sulfonamide (**P7**), and phosphines with an amide function (**P8** & **P9**) were all ineffective, yielding addition products with low *ee* values (Table 1, entries 7–9). Thiourea-containing **P10** turned out to be a good catalyst, almost comparable to the corresponding carbamate-bearing **P4** (Table 1, entry 10 vs. entry 4). Mono-functional, C_2-symmetric **P11** was effective in



Scheme 1. (Color online) Enantioselective synthesis of mixed 3,3′-bisindole core structures from 3′-indolyl-3-oxindoles via a phosphine-catalyzed umpolung γ-addition.

Table 1
Reaction optimization[a].



Entry	Catalyst	Solvent	*T* (°C)	% 3a[b]	% *ee*[c]
1	**P1**	Toluene	rt	93	88
2	**P2**	Toluene	rt	99	84
3	**P3**	Toluene	rt	99	83
4	**P4**	Toluene	rt	96	80
5	**P5**	Toluene	rt	87	75
6	**P6**	Toluene	rt	70	7
7	**P7**	Toluene	rt	94	39
8	**P8**	Toluene	rt	84	27
9	**P9**	Toluene	rt	93	21
10	**P10**	Toluene	rt	94	79
11	**P11**	Toluene	rt	95	31
12	**P1**	CH_2Cl_2	rt	88	76
13	**P1**	THF	rt	95	75
14	**P1**	Et_2O	rt	89	82
15	**P1**	CH_3CN	rt	31	61
16	**P1**	Toluene	0	97	94
17	**P1**	Toluene	−10	98	96
18	**P1**	Toluene	−20	84	93
19[d]	**P1**	Toluene	−10	90	91
20[e]	**P1**	Toluene	rt	90	74
21[f]	**P1**	Toluene	rt	95	87

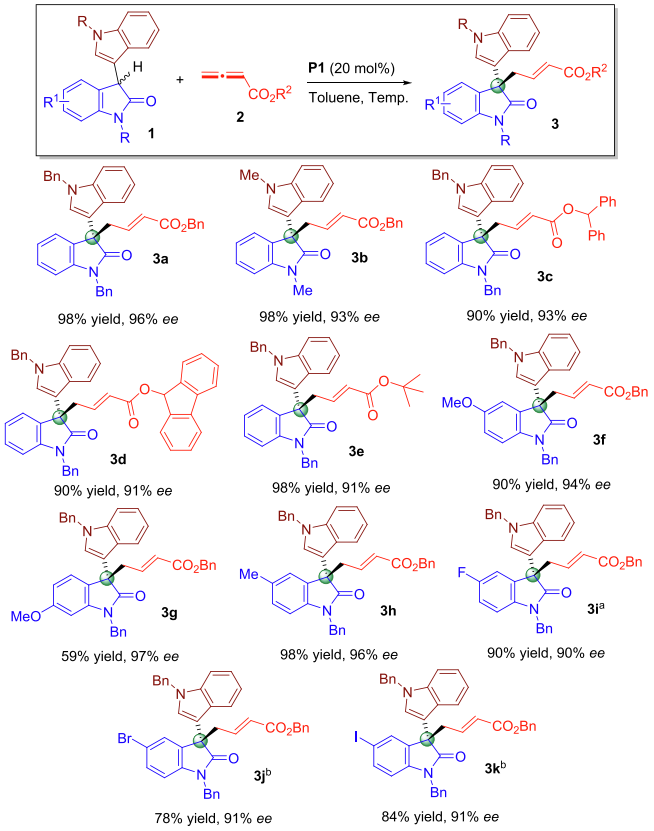
[a] Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), and the catalyst (0.02 mmol) in the solvent specified (1 mL) at indicated temperatures.
[b] Yields refer to isolated yields.
[c] The *ee* values were determined by HPLC analysis on a chiral stationary phase.
[d] The catalyst loading was 10 mol%.
[e] PhOH (20 mol%) was used as an additive.
[f] Na_2CO_3 (20 mol%) was used as an additive. Bn: benzyl; rt: room temperature.

promoting the reaction, but ineffective in asymmetric induction (entry 11). Accordingly, OTBS-L-Thr-based **P1** was identified as the optimal catalyst, and subsequent solvent screening confirmed that toluene is the solvent of choice (Table 1, entries 12–15). Further enhancement was achieved by carrying out the reaction at low temperatures (Table 1, entries 16–18). The effects of reducing the catalyst loading and inclusion of different additives were also explored (Table 1, entries 19–21). Under the optimized reaction conditions, **P1** effectively catalyzed the γ-addition of 3′-indolyl-3-oxindole **1a** to allenoate **2a**, affording the desired product **3a** in 98% yield and with 96% *ee* (Table 1, entry 17).

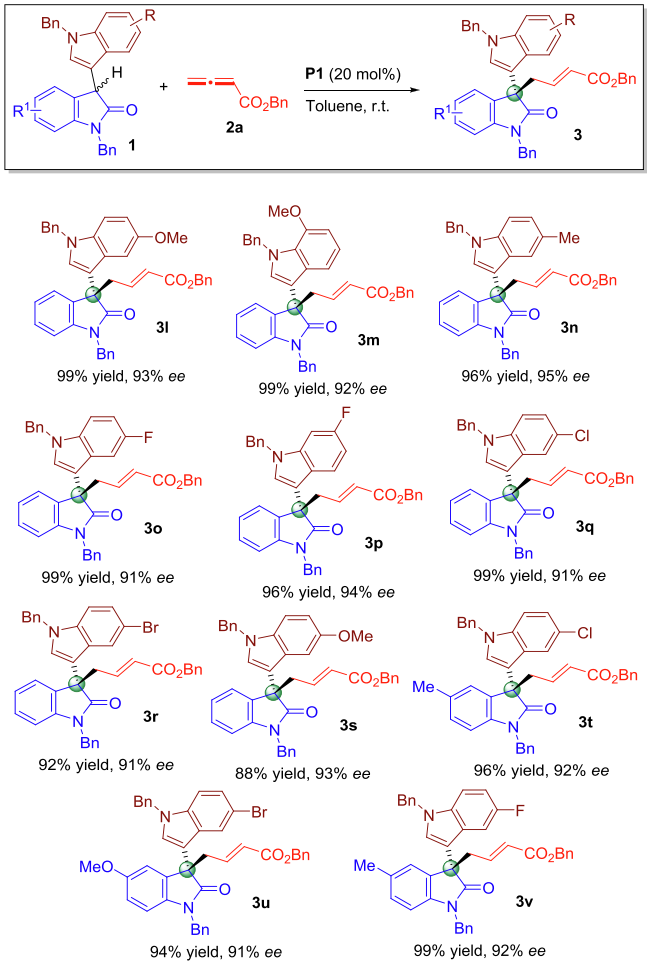
With the optimized reaction conditions in hand, the substrate scope of the reaction was investigated. In general, both oxindole moieties (**1**) and allenoate structures (**2**) could be varied, and the resulting mixed 3,3′-bisindoles **3** were attainable in mostly good yields and with excellent enantioselectivities (Scheme 2). 3′-Indolyl-3-oxindole bearing either an *N*-benzyl group or an *N*-methyl substituent gave equally good results (**3a** and **3b**). The reaction tolerated well the employment of allenoates **2** with different ester groups (**3c**, **3d**, and **3e**). While 3′-indolyl-3-oxindoles **1** with an electron-rich oxindole moiety worked very well for the reaction (**3f**, **3g**, and **3h**), substrates bearing an electron-poor oxindole were less efficiently converted; lower reaction temperatures and longer reaction times were required for attaining desired results (**3i**, **3j**, and **3k**).

Next, the reaction scope with regard to the indole moieties in 3′-indolyl-3-oxindoles **1** was investigated (Scheme 3). The reaction was applicable to a wide range of **1** bearing different indole subunits, regardless of their electronic nature and substitution patterns, and consistently high yields and excellent *ee* values were attainable (**3l** to **3s**). Furthermore, equally good results were obtained when both oxindole and indole substructures in **1** were varied at the same time (**3t**, **3u**, and **3v**).

We have not investigated the reaction mechanism in detail at this stage but do believe that the reaction disclosed herein follows the general mechanistic pathways described in the literature



Scheme 2. (Color online) Substrate scope. Reaction conditions: **1** (0.1 mmol), **2** (0.12 mmol), and **P1** (0.02 mmol) in toluene (1 mL) at −10 °C. Yields refer to isolated yields; the *ee* values were determined by HPLC analysis on a chiral stationary phase. [a]The reaction was carried out at −40 °C for 48 h. [b]The reaction was carried out at −65 °C for 24 h. Bn: benzyl.



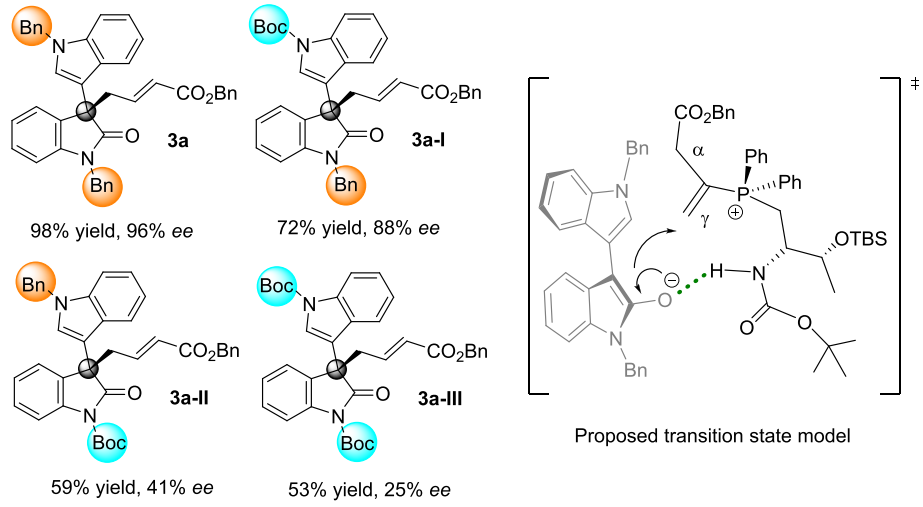
Scheme 3. (Color online) Further substrate scope. Reaction conditions: **1** (0.1 mmol), **2a** (0.12 mmol), and **P1** (0.02 mmol) in toluene (1 mL) at room temperature. Yields refer to isolated yields; the *ee* values were determined by HPLC analysis on a chiral stationary phase. Bn: benzyl.

[55,62]. A few experiments were performed to gain some preliminary understanding of asymmetric induction. We prepared 3′-indolyl-3-oxindoles **1** bearing different protective groups, i.e., benzyl and Boc, at the oxindole and indole nitrogen positions, and studied their γ-addition reactions to allenoate **2a** under otherwise identical conditions (Scheme 4). Whereas employment of the optimal substrate (**1a**) bearing two benzyl groups at both oxindole and indole nitrogen atoms furnished **3a** in 98% yield and 96% *ee*, replacing either or both benzyl groups with Boc protection was detrimental. Installing a Boc group at indole nitrogen position led to the formation of **3a-I** in 72% yield and 88% *ee*. Replacing the oxindole benzyl with a Boc drastically lowered the enantioselectivity (**3a-II**, 59% yield and 41% *ee*), and the situation became worse (**3a-III**, 53% yield and 25% *ee*) when both benzyl groups were replaced. The above preliminary experimental findings suggest that the hydrogen bonding interaction between the carbamate NH of the catalyst and the oxindole enolate is crucial, which facilitates the overall transformation and is key to the asymmetric induction. Different levels of disturbance to this vital hydrogen bond result in respective reduced chemical yields and decreased enantioselectivities (DFT calculations are on-going to understand the origin of observed stereoselectivities).

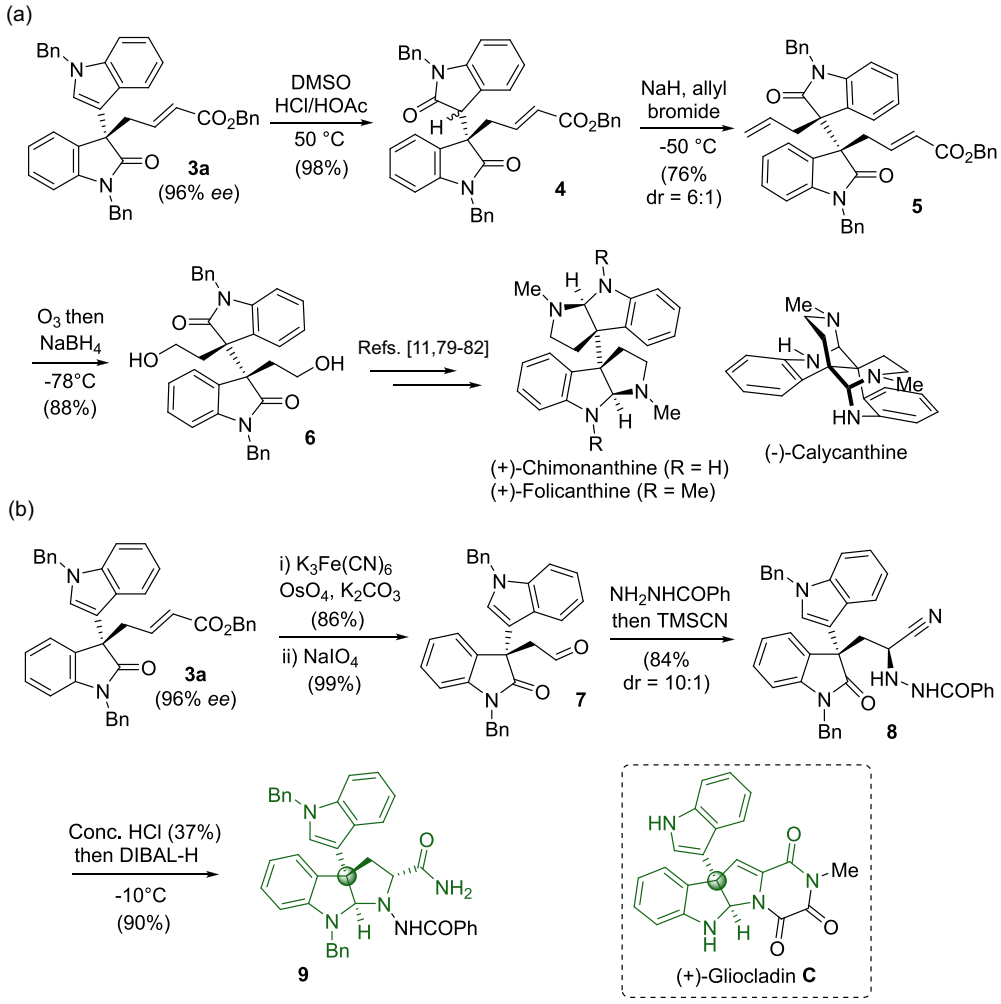
The highly enantioselective umpolung γ-addition of 3′-indolyl-3-oxindoles to allenoates disclosed herein readily creates optically enriched mixed 3,3′-bisindole scaffolds, which are highly valuable synthetic intermediates, especially for the preparation of natural products. As an illustration, the γ-addition product **3a** was oxidized to bisoxindole **4** by treating with DMSO/HCl, and subsequent allylation afforded bis-allylic **5** which contains two quaternary stereogenic centers. Ozonolysis of **5**, followed by reduction, then furnished diol **6** (the absolute configuration of compound 6 was assigned based on X-ray crystallographic analysis (CCDC 1918373)), which can be converted to (−)-Chimonanthine, (−)-Folicanthine and and (−)-Calycanthine following literature procedures [11,77–80]. Alternatively, sequential dihydroxylation-oxidation of **3a** formed aldehyde **7**, in which the indole core was retained. Compound **7** then underwent a three-component reaction with benzoyl hydrazine and TMSCN to afford α-aminonitirile **8** in a highly diastereoselective manner. Finally, hydrolysis and reductive cyclization furnished compound **9**, the core structure of Gliocladin C and its related nature product family [81–87] (Scheme 5).

4. Conclusion

In conclusion, we have developed a highly enantioselective umpolung γ-addition of 3′-indolyl-3-oxindoles to allenoates, catalyzed by amino acid-derived bifunctional phosphines. Notably, this is the first time that 3′-indolyl-3-oxindoles are used as prochiral pronucleophiles, and our strategy allows for facile creation of a variety of quaternary carbon-containing mixed 3,3′-bisindole cores, which are structurally challenging and synthetically useful.



Scheme 4. (Color online) Employment of different protective groups in 3′-indolyl-3-oxindoles.



Scheme 5. (Color online) Formal total syntheses of (+)-Chimonanthine, (+)-Folicanthine and (-)-Calycanthine and concise synthesis of pyrrolidinoindoline core structure.

Synthetic values of γ-addition products and the power of our methodology are demonstrated in concise formal total syntheses of (−)-Chimonanthine, (−)-Folicanthine and (−)-Calycanthine, as well as facile construction of core structure of pyrrolidinoinoline. Currently, we are working towards developing more phosphine-catalyzed asymmetric processes for the total syntheses of a broad range of bisindoline alkaloids and their structural analogues, and evaluating their biological activities.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

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Author contributions

Yildiz Tasdan and Guang-Jian Mei performed and analyzed the experiments. Guang-Jian Mei and Yixin Lu conceived and designed the project. Yixin Lu overall supervised the project. Yildiz Tasdan, Guang-Jian Mei and Yixin Lu prepared this manuscript.

Appendix A. Supplementary materials

Supplementary materials to this article can be found online at https://doi.org/10.1016/j.scib.2020.01.017.

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